

Fig. 2: Actual-size PCB of the device for charging three batteries

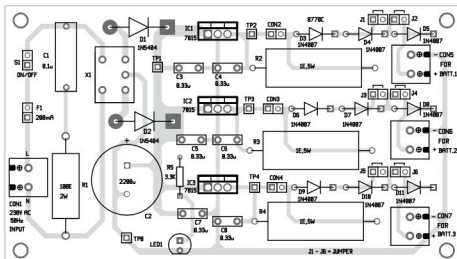


Fig. 3: Component layout of the PCB

common heat-sink having thermal resistance below 4°C/W. LED1 is used as a power on/off indicator.

Most 12V, 7Ah batteries can be connected for a long time but should be kept in charged state. The only adjustment that needs to be made to the device is to open or close jumpers J1 to J6 according to charging voltage requirement.

When jumper J1 and J2 are opened, charging voltage for battery (BATT.1) will be around 15V-0.7V = 14.3V. If J1 is opened and J2 is closed, charging voltage for battery (BATT.1) will be $(15V-0.7V) \times 2 = 13.6V$. Similarly, if both jumpers J1 and J2 are closed, charging voltage for battery BATT.1 will be $(15V-0.7V) \times 3 = 12.9V$.

PARTS LIST

Semiconductors:

- IC1-IC3 - 7815, 15V voltage regulator with heat-sink
- D1, D2 - 1N5404 rectifier diode
- D3-D11 - 1N4007 rectifier diode
- LED1 - 5mm LED

Resistors (all 1/4-watt, ±5% carbon, unless stated otherwise):

- R1 - 100-ohm, 2W
- R2-R4 - 1-ohm, 5W
- R5 - 3.3-kilo-ohm

Capacitors:

- C1 - 0.1µF, 630V polyester
- C2 - 2200µF, 40V electrolytic
- C3-C8 - 0.33µF ceramic disk

Miscellaneous:

- BATT.1-BATT.3 - 12V, 7Ah battery for charging with connector terminals
- S1 - On/off switch
- F1 - 200mA fuse with holder
- CON1 - 2-pin connector terminal
- CON2-CON4 - 2-pin connector terminal
- CON5-CON7 - 2-pin connector terminal
- X1 - 230V AC primary to 15V-0-15V, 3A secondary transformer
- J1-J6 - 2-pin jumper connector

The same will be applicable for other jumper such as J3, J4 and J5, J6 for BATT.2 and BATT.3, respectively.

Construction and testing

An actual-size, single-side PCB of the device for charging 12V batteries is shown in Fig. 2 and its component layout in Fig. 3. Enclose the PCB in a small box such that 230V AC mains can be connected easily.

Also, fix three terminals on the rear side of the cabinet for connecting the batteries.

Verify the test points as per the table before using the circuit. **EFY**



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